WASP-36b

R-1702

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_200217` · created 2026-08-02T14:18:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458896.7029 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.049
Transit depth (Rp/Rs)^2	2.6 +/- 1.59 [%]
Orbital Inclination (inc)	87.93 +/- 2.83
Airmass coefficient 1 (a1)	3.51 +/- 1.84
Airmass coefficient 2 (a2)	-0.69 +/- 0.28
Scatter in the residuals of the lightcurve fit is	52.31 %
Best Comparison Star	#1 - [409, 195]
Optimal Aperture	2.88
Optimal Annulus	3.75
Transit Duration (day)	0.0925 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.049 vs 0.1368 (deviation 0.49σ)
Duration fit vs published	0.0925 d vs 0.0758 d (deviation 1.79σ)
Mid-time O–C	8.4 min
Depth SNR	3.3
Residual scatter	52.31 %

Light curve

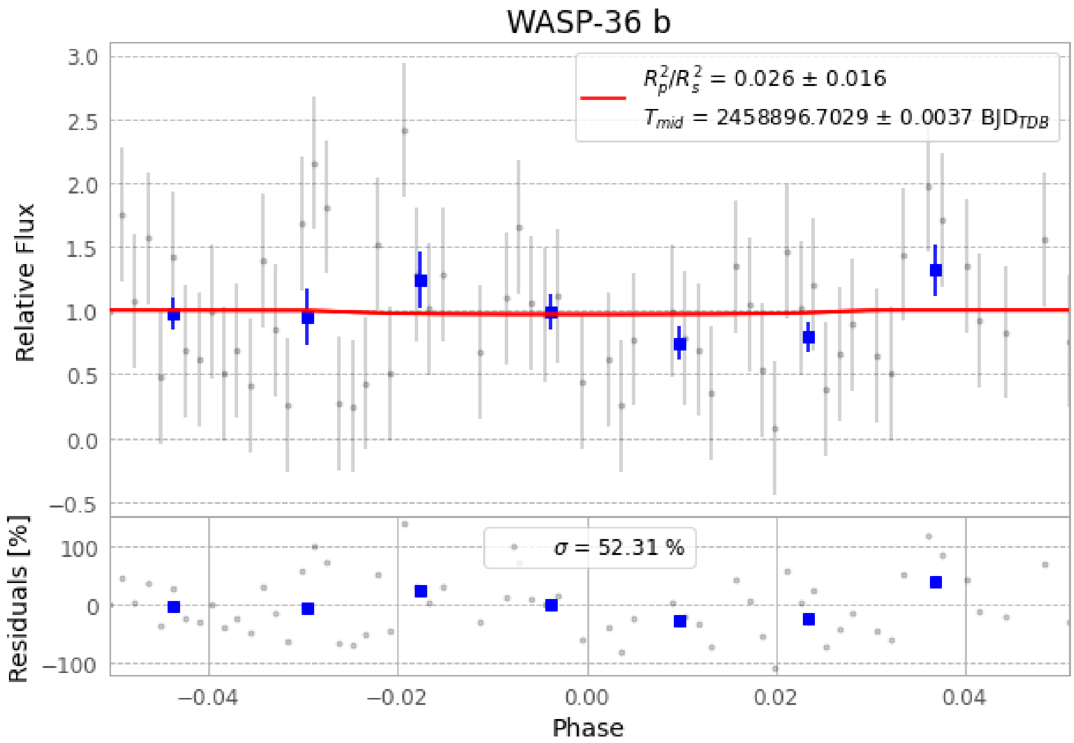

Figure 1 — FinalLightCurve_WASP-36 b_2020-02-16.png (SHA-256 2a19bd0615233909…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 396], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 26686a6a3b5f739c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

75 FITS frames (49 MB), `WASP-36200217024812.FITS` → `WASP-36200217063612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-200217.log (6a60f1645536…), FinalLightCurve_WASP-36 b_2020-02-16.png (2a19bd061523…), FinalLightCurve_WASP-36 b_2020-02-16.csv (595dd6b7db12…)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)